Addition Reactions

Exam code: 0620 & 0971

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Addition reactions

Extended tier only

- Alkenes undergo **addition reactions**, where atoms from a small molecule add across the C=C double bond
- In an addition reaction:
  - The C=C bond breaks
  - New single bonds form
  - Only one product is formed

Alkenes and bromine

- The reaction between an alkene and bromine is an addition reaction.
  - Bromine atoms add across the C=C double bond
  - A **saturated compound** is formed
- Aqueous bromine is orange and is used as a test for alkenes.
  - When bromine water is added to an **alkene**, it is **decolourised** (orange → colourless)
  - This happens because bromine reacts with the C=C bond and is no longer in solution
- Example:

Stuck on something?

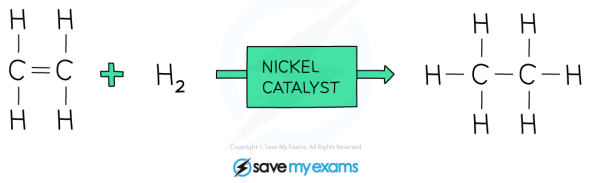
Ask questions and get help with exactly what you're revising.

Got it

Bromine atoms add across the C=C in the addition reaction of ethene and bromine

Alkenes and hydrogen

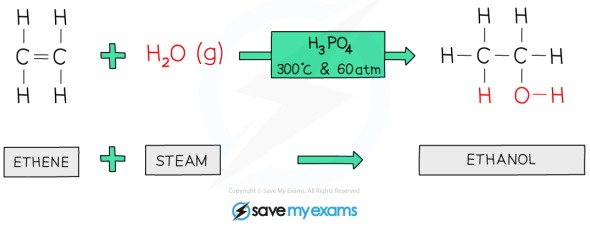
- When alkenes react with hydrogen, an **alkane** is formed
  - This is an addition reaction
  - A **nickel catalyst** is required
- Examples:
  - Ethene + hydrogen → ethane
  - Propene + hydrogen → propane



Hydrogen atoms add across the C=C in the addition reaction of ethene and hydrogen

Alkenes and steam

- When alkenes react with steam, an **alcohol** is formed
  - This is an addition reaction
  - It is also called a **hydration reaction**
  - An **acid catalyst** is required
- Examples:
  - Ethene + steam → ethanol
  - Propene + steam → propanol



A water molecule adds across the C=C in the addition reaction (hydration) of ethene with water to produce ethanol

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